

Information Technologies for Industrial Engineers

เทคโนโลยีสารสนเทศสำหรับวิศวกรอุตสาหกรรม

Smart contract applications

Decentralized lottery

Decentralized lottery

- Code
- Unit converter

ERC Token

Example: LINK (Valuable)

- <https://www.bitkub.com/th/market/link>

Example: LINK (No Value)

- Sepolia Testnet

0x779877A7B0D9E8603169DdbD7836e478b4624789

- Base Sepolia

0xE4aB69C077896252FAFBD49EFD26B5D171A32410

- Polygon Amoy

0x0Fd9e8d3aF1aaee056EB9e802c3A762a667b1904

- Arbitrum Sepolia Testnet

0xb1D4538B4571d411F07960EF2838Ce337FE1E80E

What is a token?

- Something of value
 - Currency
 - Voting right
 - Stock
- Token standard
 - EIP (*Ethereum Improvement Proposal*)
 - Guideline
 - ERC (*Ethereum Request for Comments*)
 - Implementation

Popular token standards

- **ERC-20**
 - Fungible tokens
 - Most used for representing currency
- **ERC-721**
 - Non-fungible tokens (NFTs)
 - Most used for representing digital artwork and collectibles
- **ERC-1155**
 - Multi-token standard
 - Combining the abilities of ERC-20 and ERC-720

Timeline

Source

Token list

- <https://etherscan.io/tokens>

Let's make your own ERC-20 token.

- [Contract generator](#)
- [Source code](#)

ERC20

ERC721

ERC1155

Stablecoin*

Real-World Asset*

Account

Governor

Custom

SETTINGS

Name

MyToken

Symbol

MTK

Premint

1000

FEATURES

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> ?

```
// SPDX-License-Identifier: MIT
// Compatible with OpenZeppelin Contracts ^5.4.0
pragma solidity ^0.8.27;

import {ERC20} from "@openzeppelin/contracts/token/ERC20/ERC20.sol";
import {ERC20Permit} from "@openzeppelin/contracts/token/ERC20/extensions/ERC20Permit.sol";

contract MyToken is ERC20, ERC20Permit {
    constructor(address recipient)
        ERC20("MyToken", "MTK")
        ERC20Permit("MyToken")
    {
        _mint(recipient, 1000 * 10 ** decimals());
    }
}
```